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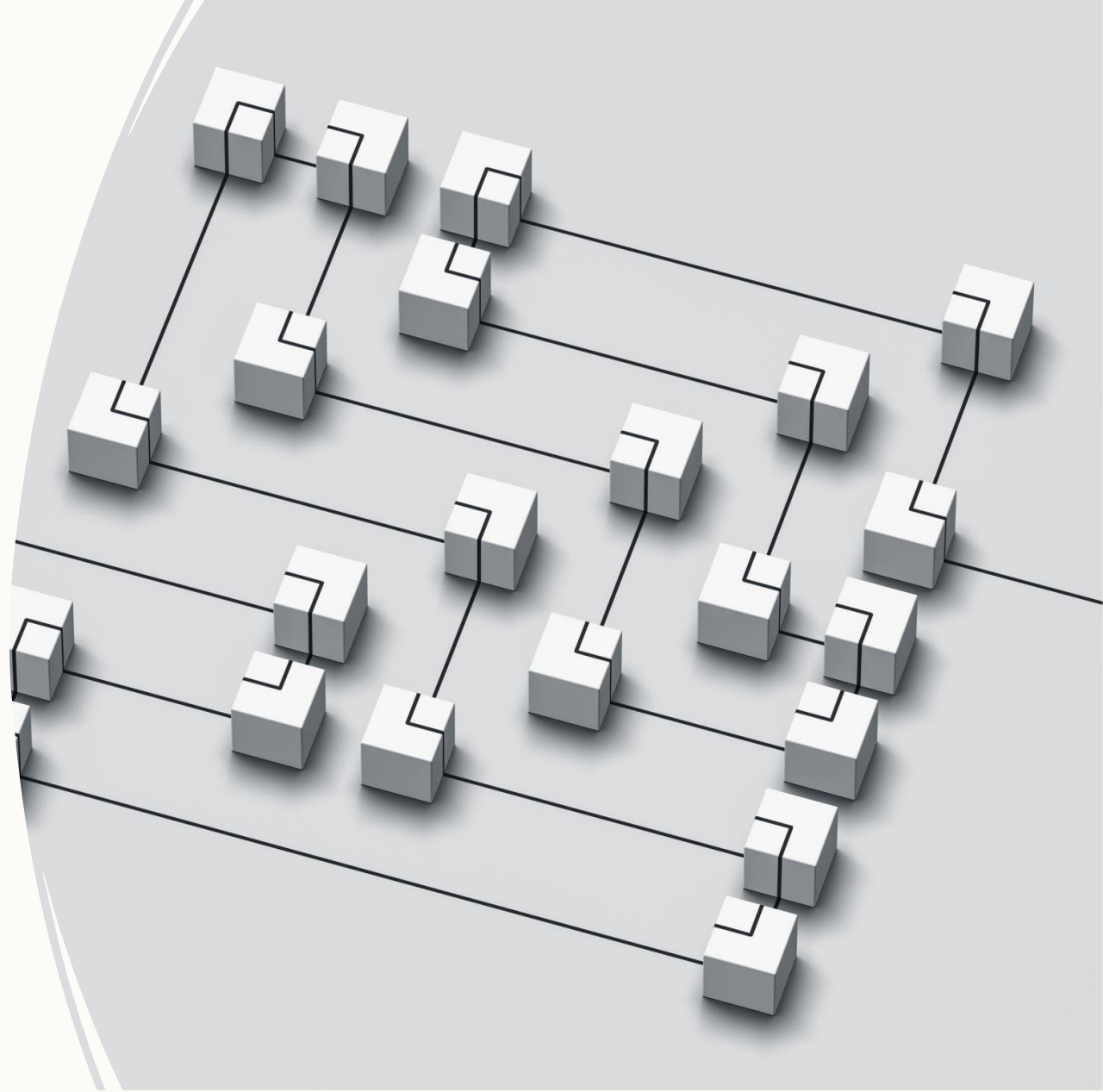
COMP 3056 Professional Ethics in Computing

Week 8 Intellectual Property



Learning outcomes

- Intellectual & Intangible Property
- Copyright
- Patent Law
- Trademark
- Trade Secrets
- IP Protection for Software





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Intellectual & Intangible Property

Intellectual & Intangible Property

创作和发明的成果（例如一本书、一首乐曲、计算机软件……）通常被称为“知识产权”，以将其与有形财产（例如房屋、汽车、一双鞋……）区分开来。

- The **products of creation and invention** (e.g., a book, a musical composition, computer software...) **are typically called “*intellectual*” property** to distinguish them from ***tangible*** property (e.g., a house, a car, a pair of shoes...)
- Today, we look at:
 - 受计算机和互联网影响最大的知识产权类型
Types of intellectual property that are most affected by ***computing and the Internet***
 - 由于计算机而产生的新类型无形财产，但似乎不具有创造性性质
New types of intangible property that arise due to computing, but ***do not seem creative*** in nature

知识产权 = 保护“无形创作成果”的法律体系。数字时代让这些保护更复杂，软件、数据、算法等新型无形资产不断出现，而它们未必适合传统的版权 / 专利方式，因此需要重新思考如何保护创新。

Additional Reading

- Chapter 4 of the book Ethics in a Computing Culture (Brinkman and Sanders, 2013)
- Chapter 4 of the book A Gift of Fire (Baase, 2013)
- Chapter 3 of the book Ethics for the Information Age (Quinn, 2013)
- Chapter 8 of the book Ethics and Computing (Bowyer, 2001)
- Legally Speaking Mass Digitization as Fair Use. Pamela Samuelson, Communications of the ACM, Vol. 57, No. 3, pp. 20-22, March 2014.
- Does the Patent Office Respect the Software Community? Gregory Aharonian, IEEE Software, pp. 87-89, July/August 1999.

The basic idea / argument...

创造力和发明的重要性：

这一点强调：

社会的进步来自于 creativity（创造力）和 invention（发明）。

如果没有人写书、作曲、发明软件或技术，我们不会有知识增长与技术进步。

- The importance of creativity and invention

保护创新者及其创作

– *Protecting innovators and their creation*

最终的目的是鼓励创新

- Ultimately to encourage innovation

但是，是否“反而阻碍创新”呢

– *But does the opposite happen?*

- Defining *intellectual* and *intangible* property

– Intellectual property often shortened to IP

知识产权制度的基本理念是：保护创新者，使其能够从自己的创作中受益，从而在社会中激励更多创新。

然而，现实中知识产权法也可能产生反效果：通过过度保护、长期垄断或技术性控制（如 DRM），反而限制了他人的创造与发展。因此，在理解 IP 时，必须同时看到其“保护创新”与“可能阻碍创新”的双面性。

(Intellectual) Property of Ideas

“If nature has made any one thing less susceptible than all others of exclusive property, it is the action of the thinking power called an idea, which an individual may exclusively possess as long as he keeps it to himself; but the moment it is divulged, it forces itself into the possession of everyone, and the receiver cannot dispossess himself of it.”

[Thomas Jefferson, 1813]

President, Inventor, and Founding Father of USA

- hailed as the author of the *Declaration of Independence*

Intellectual Property (IP) Law

确保有好点子的人能够从中获利：给创作者提供经济激励

- To ensure that people with **good ideas** can **profit** from them

即使这个想法已经为所有人所知：

— **even once the idea is known to everyone**

即使一个 idea 被公开（比如你发明了微信支付，或设计了某个算法），版权 / 专利法仍然允许你对它享有一定期限的独占权益。这是在“自然属性”（ideas 天然不可独占）上加的人工保护。

- To **encourage** people to get involved in **creative arts** and **invention** 鼓励人们投入到艺术创作和发明活动中：让更多人愿意创作作品、发明技术

— **for the benefit of everyone** 最终是为了整个社会的好

- IP Law is very **complicated...** 复杂

— *I know this first hand!*

- We focus on **three key aspects** that directly affect computing

1. **Copyright** 版权

2. **Patent** 专利

3. **Trademark** 商标

IP Protection: 5 Key Questions

- For each type of IP protection, we ask:

1. 什么样的东西可以受到保护？

1. *What type of thing can be protected?*

2. 作品的创作者保留哪些权利

2. *What rights are reserved for the creator of the work?*

3. 公众保留哪些权利？

3. *What rights are reserved for the public?*

4. 如何为作品获取保护？

4. *How does one obtain protection for a work?*

5. 这种保护能持续多久？

5. *How long does the protection last?*

The Numbers: China

- China: Intellectual Property Rights - Protecting assets in the information, communications and entertainment market. The Economist, HK, 2005
<https://www.kpmg.com/CN/en/IssuesAndInsights/ArticlesPublications/Documents/China-Intellectual-Property-Rights-200502.pdf>
- The added value of China's patent-intensive industries topped 10.7 trillion yuan (\$1.51 trillion) in 2018, contributing 11.6 percent to the country's GDP (reported 2020).
https://english.www.gov.cn/statecouncil/ministries/202003/26/content_WS5e7c056cc6d0c201c2cbf7bd.html
- International integration:
[THE 5 BIGGEST CHINA IP STORIES OF 2021 | Insights | CBBC](#)

HOMEWORK: what is the current situation?



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Copyright

Copyright

- Main mechanism for **protecting creative works**, e.g., **art, music, & literature**
- US Law **DOES PROTECT:**
原创的作者作品，需要固定在有形媒介中：文学、音乐、戏剧、哑剧、平面艺术、雕塑、电影、录音和建筑作品
 - “**Original works of authorship fixed in any tangible medium of expression**” (Title 17, US Code section 102) in the areas of literature, music, drama, pantomime, graphic art, sculpture, motion pictures, sound recordings, and architecture
- US Law **DOES NOT PROTECT:**
想法，事实或常识
 - Ideas, facts, or common knowledge
有创意但是没有固定形式的作品
 - Creative works that are not in a “tangible” fixed form
 - E.g., dance choreography: to protect work, choreographers often make video recordings of dance during rehearsal
舞蹈作品只有被录影、写成脚本、画成舞谱后才受保护因为它必须 “fixed in a tangible medium”
这说明：版权保护“表达”，不保护“未固定的创意”。
- No “**process**” to obtain copyright – works created are **automatically protected once they appear in a fixed medium** 无需“流程”即可获得版权——作品一旦以固定形式呈现即自动受保护
- Duration – (Currently) Lifetime of the author + 70 years
版权期限：作者生前 + 70 年
 - When Old, Out-of-Copyright Art Becomes Free Art
老作品一旦版权到期，就成为自由艺术

版权赋予作者的四项主要的权利

Four main rights for the author

复制权：作者拥有复制作品的专有权利

1. The right to reproduce the copyrighted work

衍生作品权/改编权：作者拥有基于原作品创作“衍生作品”的权利。

2. The right to prepare derivative works based on the copyrighted work

发行权：作者控制作品“传播给公众”的方式与条件。包括：销售（sell）出租（rental）借出（lending）转让所有权（transfer of ownership）；数字内容的分发属于 distribution！

3. The right to distribute copies of the copyrighted work to the public by sale or other transfer of ownership, or by rental, lease, or lending

公开表演/公开展示权：作者拥有公开演示、播放、展览作品的权利。

4. The right to perform or display the copyrighted work publicly

权利	关键词	例子
1 复制权	reproduce	下载、复制、扫描
2 衍生作品权	derivative works	改编、翻译、修改代码
3 发行权	distribute	出售、出租、上传到网站
4 公开展示/表演权	perform/display publicly	播放电影、展示图像、直播放音乐

Rights for the public

所有不被作者的四项专有权覆盖的行为，都属于公众权利。

- Most activities **not covered** by four main **rights of the author** belong to the public.
 - E.g., if you own a DVD, you have right to **display privately in own home**
- There are **two main rights** reserved **for public** that deserve special attention:
 1. **Fair Use** 合理使用
 2. **First Sale** 首售原则

受版权保护的作品不等于公众毫无权利；公众也有部分保留权利。

Fair Use

在某些情况下，我们可以在未经明确许可的情况下使用受版权保护作品的部分内容
这是 Fair Use 的核心定义：

即使作品受版权保护，只要符合特定条件，公众可以合法使用其中一部分，而不构成侵权。

- We can, in certain situations, use parts of copyrighted work without express permission

以下用途不属于版权侵权

- Following are **not** an infringement of copyright:

- Criticism 批评：你可以引用作品内容进行批评或评审。
- Comment 评论：比 criticism 更广，指加入个人观点、分析或讨论。
- News Reporting 新闻报道：媒体引用内容用于报道属于 Fair Use。
- Teaching (inc. multiple copies for classroom)
- Research 研究：学术研究可引用作品，用于分析、比较、讨论等。

教学，包括课堂分发多份复印件
注意：

“合理数量”很重要。
整本教材复印全班复印 = 侵权
复印一章 = 边界案例，要参考
4-factor test
复印几页 = 通常属于 fair use

Fair Use 存在的理由是为了：确保教育、批评、研究、新闻等公共利益活动不被版权制度过度限制。
它是保护公众的重要机制，是版权制度“平衡创作者与公众”保留权的核心。

Factors of Fair Use

合理使用的判断因素

- Law does not give clear rules 法律没有给出明确的规则
法院/仲裁者会根据四个因素判断公平使用
- Four factors considered by court /arbitrator for each case:
 1. Purpose and character of use 用途和使用性质
 - e.g., commercial vs. non-profit educational 商业用途 还是 非盈利教育用途
商业 (commercial) = 不利于 Fair Use 教育 / 非盈利 (non-profit) = 有利于 Fair Use
 - noncommercial and educational more likely to be fair
 2. Nature of the copyrighted work 受版权保护作品的性质
 - e.g., fiction vs. nonfiction, published vs. non-published 虚构/非虚构/已出版/未出版
使用非虚构作品和已发表作品更有可能构成合理使用
 - use of nonfiction work & published work more likely to be fair
 3. Amount and substantiality of used material 使用量和实质性
 - relative to the copyrighted work as a whole 重点是“相对”数量，而不是绝对量
短小的摘录更有可能是合理的，但如果实质内容较多则不
 - short extracts more likely to be fair, but not if substantial
使用对版权作品潜在市场或价值的影响
 4. Effect of use on potential market for, or value of, the copyrighted work
 - use more likely fair if copyright holder's ability to make money is not damaged
如果你使用该内容不会减少版权人赚钱的能力，更可能被认为是 Fair Use。

不属于合理使用：尽管苹果和脸只是原作的一小部分，但它却是至关重要的或可识别的部分，因此属于实质性部分。

Not fair use: although apple and face is a small part of original work, it is a vital or recognisable part, and therefore *substantial*



The Son of Man (1964), René Magritte

https://en.wikipedia.org/wiki/The_Son_of_Man

Scenario

- Prof. Smith wants to provide students with a chapter of a textbook (not authored by him) as supplementary material on a module. Smith posts a PDF file of the chapter and posts it in a public website. **Qu: Is this fair use?** NO：发到公平平台整本书，本质上是免费向互联网发布受版权保护的完整章节
- What if Smith places on a password-protected site for students only? **Qu: Is this fair use?** maybe YES：教育目的，而且非虚构但整本书数量有点过多
- What if Smith takes a chapter from one book, another chapter from another book, etc. and posts all on a password-protected site for students. **Qu: Is this fair use?** maybe NO：虽然教育目的且非虚构，数量也有限；但多本书各一章合起来等价于使用量巨大，接近于侵权

To consider: Purpose of use (education), type of work (non-fiction), amount / substantiality, effect on market/ value

Doctrine of First Sale

首次销售原则

- Copyright Law: 版权保护的是“无形作品”，即创作者的表达
 - Designed to protect something *intangible* (i.e. an *expression* of creativity)
但过去，人们购买版权作品的方式是购买“实体物品”
 - Until recently, copyrighted material was purchased in *tangible* form
 - 听音乐的权利：购买实体唱片或 CD
Right to listen to music recording: purchase physical record, or CD
 - 阅读小说的权利：购买印刷版书籍
Right to read a novel: purchase printed copy of the book
 - This creates confusion in minds of consumers—it's natural to think same set of rules (about resale and copying) apply to CDs and books as to shovels, **but that is not the case...**
- Consider a textbook you have purchased
 - If you purchased a new copy, then the authors (copyright holders) received a royalty of what was paid
 - Now, **if you sell it back to bookstore and they resell it as a used book, do authors get a second royalty?**

设想一下你买过的一本教科书：

——如果你买的是新书，那么作者（版权所有人）会收到你所支付金额的版税；
——那么，如果你把它卖回书店，书店再把它作为二手书出售，作者会收到第二次版税吗？
首次销售原则：版权保护内容，实体副本属于消费者。你可以买卖书本，但不能复制内容。

Doctrine of first sale: No second royalty

- Copyright holders are **not** entitled to a second royalty from resale 版权持有人没有权利从二手转售中获得第二次版税
 - “*The owner of a particular copy ... lawfully made under this title ... Is entitled, without the authority of the copyright owner, to sell or otherwise dispose of the possessions of that copy*”
有权在未经版权所有者授权的情况下出售或以其他方式处置该复制品”
 - i.e., you are free to sell your book without involving the copyright owner **as long as no new copy is made**
也就是说，只要不制作新的复制品，您就可以自由出售您的书籍，无需涉及版权所有者
 - This policy (created 1908) was workable ...
 - until books shifted to e-books and CDs shifted to MP3s
 - then, first sale may no longer make sense
 - The reason? Modern copyright protection schemes, collectively called “digital rights management”
原因在于现代版权保护方案，统称为“数字版权管理”

首售原则：可以买卖实体，不可复制内容。数字时代没有“实体副本”，DRM 阻止复制 首售原则几乎失效。

数字版权管理：DRM 技术锁定数字内容，使公众无法行使合理使用（Fair Use）和首售权（First Sale），从而削弱了数字时代的版权平衡。

Digital Rights Management (DRM)

如今许多消费者在网上购买音乐、小说等电子内容：

- 您不再收到实体产品

- 您获得的是可在不同设备上播放的虚拟电脑文件

- Many consumers now buy music, novels, electronically on the Internet
 - You no longer receive physical product
 - You get a virtual computer file that can be played on different devices
- Digital Right Management (DRM):
 - Technologies that work to ensure copyrighted content can only be viewed by the purchaser 旨在确保受版权保护的内容只能由购买者查看的技术
 - When you purchase content it is locked to your device
 - ebook purchased for your Amazon Kindle cannot be read on your friend's Kindle
- What happens to Fair Use and Fair Sale rights? (e-book with DRM)
 - You may not be able to cut and paste a quote from a book [can't exercise right to fair use] 您可能无法从书中复制粘贴引文[无法行使合理使用权]
 - You cannot give copy to friend and then delete your copy [can't exercise right to resell a book – i.e., doesn't meet first sale doctrine]
您不能将副本给朋友然后删除自己的副本[无法行使转售书籍的权利，即不符合首次销售原则]

Digital Millenium Copyright Act (DMCA)

998 年通过，用来处理现代版权问题。

- Law updated to deal with modern copyright issues (1998)

任何人不得规避保护作品访问的技术措施。

No person shall circumvent a technological measure that effectively controls access to a work protected under this title

DMCA 规定：禁止绕过 DRM（数字版权管理技术）

(Title 17, US Code, section 1201)

- Law has been criticised on grounds that it *makes fair use and fair sale legally impossible* DMCA 使 Fair Use 和 First Sale 在数字时代几乎法律上不可能实现
- Copyright law gives public rights to fair use and fair sale...
 - But can only be exercised by breaking DRM system
 - And breaking DRM is illegal under DMCA

版权法赋予公众合理使用和合理销售的权利……

——但行使这些权利必须破解数字版权管理（DRM）系统

——而根据《数字千年版权法》（DMCA），破解 DRM 是违法的

DMCA 禁止绕过 DRM，使数字内容时代的“合理使用”和“首售权”在法律上形同虚设。

Plagiarism vs. Copyright Violation

- Posner's definition of plagiarism
 - “*copying that the copier claims is original with him and the claim causes the copier's audience to behave otherwise that it would if it knew the truth*” 抄袭 = 把别人的作品说成是自己的，并让评阅者因此做出错误判断。

Scenario: Student Alice pays student Bob to write an essay on her behalf. Alice hands in the essay as her own work with the express permission of Bob.

Qu: Copyright infringement, plagiarism, both, neither?

- This is plagiarism – copier, Alice, claims it is original, and marker would discipline/fail Alice if truth was known
 - This is not copyright infringement – Bob gives Alice permission to use
- 抄袭 不等于 必然侵权
- Plagiarism does not necessarily imply copyright infringement
 - & Vice-versa: We can violate copyright by publishing a photograph on our website, even if we give credit
- 即使标明出处，也可能构成版权侵权

Sharing vs. Selling

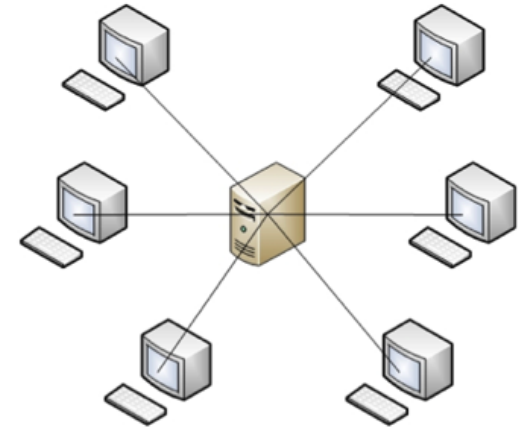
直到 1990 年代，版权盗版主要是非法复制实体介质

- Until 1990s, copyright piracy involved counterfeiting—illegal duplication of physical media
通过分享造成的侵权（例如，让朋友使用你的磁盘非法安装软件）对业务的影响相对较小（原购买者与相对较少的朋友分享）。
 - Infringement through sharing (e.g., let a friend use your disk to illegally install software)
had relatively small business impact (original purchase shared with relatively few friends)
互联网使得一份合法拷贝能够被成千上万甚至数百万人分享。
 - The internet allows a single legal copy to be shared with thousands/millions of people
例如：1994 年——麻省理工学院一名学生被指控犯有刑事版权侵权罪。
 - Example case: 1994—MIT Student charged with criminal copyright infringement
据称在其网站上免费提供受版权保护的软件（如 Excel）供人下载。
 - Allegedly hosting copyrighted software (e.g., Excel) on his website for free download
 - 判决结果：未构成版权侵犯。
 - Verdict: no copyright violation
 - 法律规定，只有当行为“出于商业利益或私人经济利益的目的”时才构成犯罪。
 - Law said criminal only if the act was “for the purpose of commercial advantage or private financial gain”
 - 没有私人或经济利益，因此侵权行为不构成犯罪。
 - No private or financial gain, therefore infringement not a criminal offense
 - Result: 1997 – No Electronic Theft Act (or NET Act)
 - Copyright law amended, so “financial gain” includes receiving copyrighted works for free
 - Now swapping (I’ll give you movie A if you give me movie B) is clearly illegal
- 结果：1997 年——《无电子盗窃法》（或 NET 法案）出台。
——版权法进行了修订，将“经济利益”定义为免费获取受版权保护的作品。
——现在，交换（我给你电影 A，你给我电影 B）显然是非法的

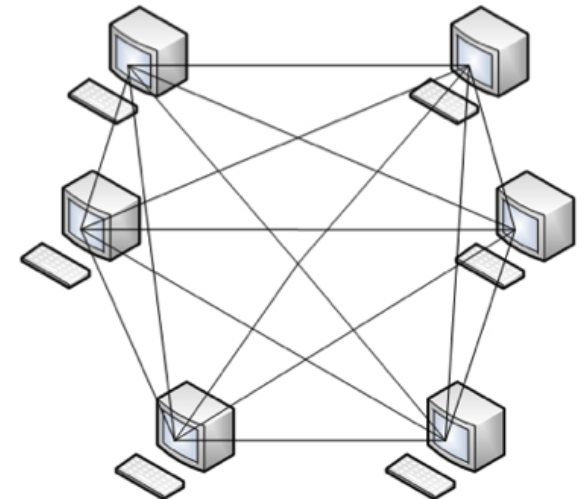
互联网让分享变成大规模侵权。NET Act 之后，即使不赚钱，只要互相交换或分享受版权保护内容，都属于刑事犯罪。

P2P Sharing & Searching

- Post a whole movie on a website?
 - Legal? Moral?
 - Probably not
- What about P2P software?
 - No central server
 - Search only – provider helps you search for files on the network – this is hard problem (would not be possible without software application)
 - No hosting from the service provider
- Legal? Moral?
- ?



A server based network opposed to peer-to-peer



A peer-to-peer network.



- The first P2P file-sharing software 第一个点对点文件分享软件
 - 80 million users at its peak 拥有8000万用户
 - College dorms: 61% of external traffic was MP3 file transfers! 61% 流量是 MP3 文件交换
- Found guilty of violating copyright law
 - Despite not hosting any copyrighted material – **correct decision?**
- Defined as “**Contributory Infringement**” 帮助侵权
 - **Infringement would not have happened without Napster’s help** 没有 Napster 的帮助，侵权行为就不会发生
- There is also “**Vicarious Infringement**” 替代责任 / 间接侵权
 - **Infringement occurred in an area under Napster’s supervision** 侵权发生在 Napster 的“控制范围内”
- Numerous examples since: The Pirate Bay, BitTorrent, etc...
 - Napster: <https://en.wikipedia.org/wiki/Napster> **Very interesting read!**
 - The Pirate Bay: https://en.wikipedia.org/wiki/The_Pirate_Bay
 - BitTorrent: <https://en.wikipedia.org/wiki/BitTorrent>

Copyright Duration

争议：版权应该保护多久？

- Debate: how long should copyright protection last
 - Term has grown steadily over time
 - Since 1998—**it is author's lifetime, plus 70 years.**
- Arguments for long term: 支持长保护期的理由
 - To support author and family for life 支持创作者及其家人
- Arguments against very long term: 反对长保护期的理由
 - Difficult to track down owner (orphaned work) 孤儿作品：无法找到版权持有人的作品
 - Social cost: students read Shakespeare at school – what would be annual cost be if works still in copyright!?
社会成本：长保护期会增加教育成本、阻碍文化发展。
公共领域 (public domain) 的作品经常是教育系统的支柱。



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Patent Law

专利保护“有用的发明”，必须申请，并且必须具有新颖性与非显而易见性，保护期 20 年，赋予发明者强大的独占权。Patents

专利

Patents: Definitions

专利是“发明创造的知识产权”，价值在于“实用性”

- Intellectual works of invention, value is on its usefulness
 - Covers processes, machines, manufacture, compositions of matter, ...
 - Capable of being made or used
- Need to apply for it 专利必须申请
与 版权自动产生 完全不同：专利不是自动拥有的，它必须申请并通过审查。此外，专利申请很贵
 - Can be expensive process (more than \$10,000)
- Criteria, invention must be: 申请条件，发明必须：
 - Novel (not been invented before) 新颖：之前从未被发明过
非显而易见的：对本领域专家来说不应是显而易见
 - Non-obvious (not obvious to another specialist in appropriate area)
- Valid for some fixed period of time 在一定期限内有效
 - (usually 20 years from application date) 大多数发明专利保护期为申请日起 20 年
- Good investment! No-one can make, use, offer to sell, or sell the invention without the patent holder's permission!

专利是非常好的商业投资：

因为法律予专利持有者垄断权 (exclusive right)：也就是：禁止别人制造 (make) 禁止别人使用 (use) 禁止别人销售 (sell) 禁止别人提供销售 (offer to sell) 禁止别人进口 (import)

Patent Wars!



Samsung vs Apple Patent Wars: Apple Wins \$120 Million Slide-to-Unlock Patent Case

Oct 2016

7 October 2016, 8:48 pm EDT By Vamien McKalin Tech Times





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Trademark

商标：商标保护品牌的“可识别元素”，包括文字、颜色、声音、形状，用于防伪及避免混淆，是品牌身份中可以法律保护的部分。

Trademarks: Definitions

合法注册一个词、短语、符号或可以识别产品/服务/公司的元素。

- Legal registration of word, phrase, symbol, or item that identifies product, service or corporation
- **Trademark vs. Brand:** 品牌是抽象概念，
商标是其中可以注册、可以法律保护的部分。
 - Brand refers to the product's identity as a whole 品牌指的是产品的整体身份
 - Trademark refers to specific element of identity elements 商标指的是身份元素中的特定部分
- Usually used to protect word or logo related to a product, so that consumers can be sure products are authentic
商标 通常用于保护与产品相关的文字或标识，以便消费者能够确定产品的真实性
- Can register trademark for identifying characters of brand, e.g.,
可以为品牌标识特征注册商标
 - **Colours:** T-Mobile holds trademark for colour magenta 颜色：T-Mobile 拥有品红颜色的商标
 - **Sounds:** Motorola holds trademark on the “Chirp” sound phones make when using direct-connection technology 声音：摩托罗拉拥有其手机使用直接连接技术时发出的“啾啾”声的商标
 - **Shapes:** Coca-cola holds trademark on “bottles, jars, or flasks with bulging, protruding or rounded sides”
形状：可口可乐拥有“瓶、罐或瓶状容器，其侧面呈凸起、突出或圆润状”的商标

Trademarks: Shapes & Colours

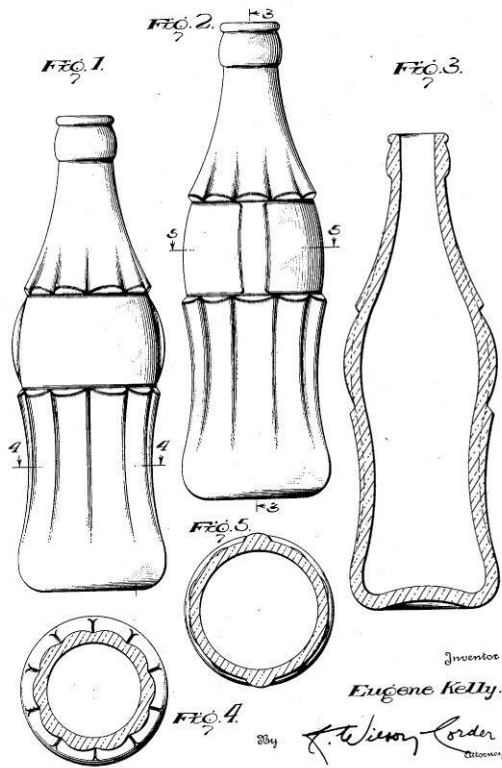
Aug. 3, 1937.

E. KELLY

Des. 105,529

BOTTLE

Filed March 24, 1937



105,529

DESIGN FOR A BOTTLE

Eugene Kelly, Toronto, Ontario, Canada, assignor to The Coca-Cola Company, Wilmington, Del., a corporation of Delaware

Application March 24, 1937, Serial No. 68,391

Term of patent 14 years

To all whom it may concern:

Be it known that I, Eugene Kelly, a citizen of the United States and resident of Toronto, Province of Ontario, Dominion of Canada, have invented a new, original, and ornamental design for a bottle, of which the following is a specification, reference being had to the accompanying drawing, forming part thereof.

Fig. 1 is a side view of my bottle.
Fig. 2 is likewise a side view, showing the bottle in Fig. 1 turned through an arc 90°, in order to better illustrate certain features of the instant design.

Fig. 3 is a vertical cross-sectional view of my bottle taken substantially along the line 3-3 of Fig. 2.

Fig. 4 is a horizontal cross-sectional view taken along the line 4-4 of Fig. 1.

Fig. 5 is likewise a horizontal cross-sectional view, taken along the line 5-5 of Fig. 2.

I claim:

The ornamental design for a bottle, as shown.

EUGENE KELLY.

The Switch

Court says T-Mobile owns the color magenta

By Brian Fung February 10, 2014

商标法保护一切可以“识别来源”的特征，不仅限于名称或 logo。包括：词语图案颜色声音形状包装三维外形（如可乐瓶）店面布局（如 Apple Store 的装修）只要它能够：识别商品来源；避免消费者混淆 就可以注册为商标。

只要某种形状或颜色足够独特，并能让消费者识别品牌，它就可以成为商标（如可口可乐瓶形、T-Mobile 的洋红色）。



(paii.hsieh)

商标必须实际使用才能注册；保护期 10 年但可无限续期；核心作用是避免消费者混淆，甚至产品的“外观风格”（trade dress）也可以受到商标保护。

Trademark

要获得商标权（TM），您必须已经在使用该标识来代表产品或服务。

- In order to receive TM, you must already be using the mark to represent product or service
可以支付几百美元的费用进行注册。
- Can register for fee of few hundred dollars
商标权有效期为 10 年，但可以无限期续展。
- TM lasts 10 years but can be renewed indefinitely
只要您拥有商标权，其他人就不能以可能引起顾客混淆的方式使用它。
- As long as you own trademark, no one else can use it in a way that is likely to cause confusion among customers
例如，其他手机公司不能采用洋红色作为公司颜色，但软饮料罐可以使用洋红色。
— E.g., no other mobile phone company can adopt magenta as company colour, but a soft-drink can could use magenta
商业外观——产品的外观和包装给人的感觉。
- **Trade dress**—look and feel of a product and its packaging
— Several companies have attempted to use trade dress protection to prevent others from copying look and feel of their Web sites
有几家公司试图利用商业外观保护来阻止他人复制其网站的外观和感觉。



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Trade Secrets

Trade Secrets

商业秘密依靠“保密 + 合同 + 法律”保护，不靠专利或版权；只要不公开，就可以永久保护，例如可口可乐配方与谷歌算法。

对于某些产品，其背后的“创意”是可以保密的

- With some products it is possible to keep the intellectual “idea” secret
 - E.g., an Internet company’s search algorithm 比如互联网公司的搜索算法
 - E.g., Coca Cola’s secret recipe 可口可乐的秘方
- Such secrets are called “trade secrets” 这类秘密被称为“商业秘密”
 - They are secrets that are fundamental to business 它们是企业赖以生存的核心机密
 - E.g., Customer delivery information for fresh chicken supplier [Faccenda]
<http://uk.practicallaw.com/books/9781845920418/chapter06#BBECCI-ch06-UID42>
 - E.g., Chemical database for mosquito net manufacturer [Vestergaard]
<https://www.rpc.co.uk/perspectives/ip/vestergaard-guarding-trade-secrets>
- IP laws are not needed to protect trade secrets 保护商业秘密无需知识产权法
 - If somebody breaks into company and steals the secret, they can be prosecuted for trespassing 如果有人闯入公司窃取秘密，可对其以非法侵入罪起诉
 - Employees can be prevented from leaking secrets through use of non-disclosure agreements in their employment contracts 通过在雇佣合同中加入保密协议，可防止员工泄露秘密
- IP laws needed only for inventions/creations that are *no longer secret*
只有当发明/创作不再是秘密时，才需要知识产权法加以保护



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IP Protection for Software

IP Protection for Software

版权、专利和商标可用于保护计算机软件产品的某些部分

- Copyright, Patents, and Trademark can be used to protect parts of a computer software product, e.g.,
 - source code logic, 源代码逻辑，源代码是作者的创造性作品，受版权保护
 - Source code is a creative work of authorship and is copyrighted
 - look and feel, e.g., 外观和感觉，例如，《吃豆人》版权——街机软件，创造性作品
 - Pac-Man Copyright — arcade software, work of creativity
 - Blue Nile Trademark — 蓝尼罗河商标——珠宝网站的商业外观法 trade dress law for jewelry website
 - features, (E.g., Apple's *Slide-To-Unlock* Patent) 功能，（例如，苹果的滑动解锁专利）
 - algorithms, (E.g., Google's *Page Rank* Patent) 算法，（例如，谷歌的网页排名专利）
- It is *not* a simple question of how to protect software
 如何保护软件并非一个简单的问题

软件可以通过版权保护代码，通过专利保护功能，通过商标保护界面风格，通过商业秘密保护算法，因此软件 IP 非常复杂，没有统一答案。

软件的安全与商标保护

Copyright & Trademark Protection for Software

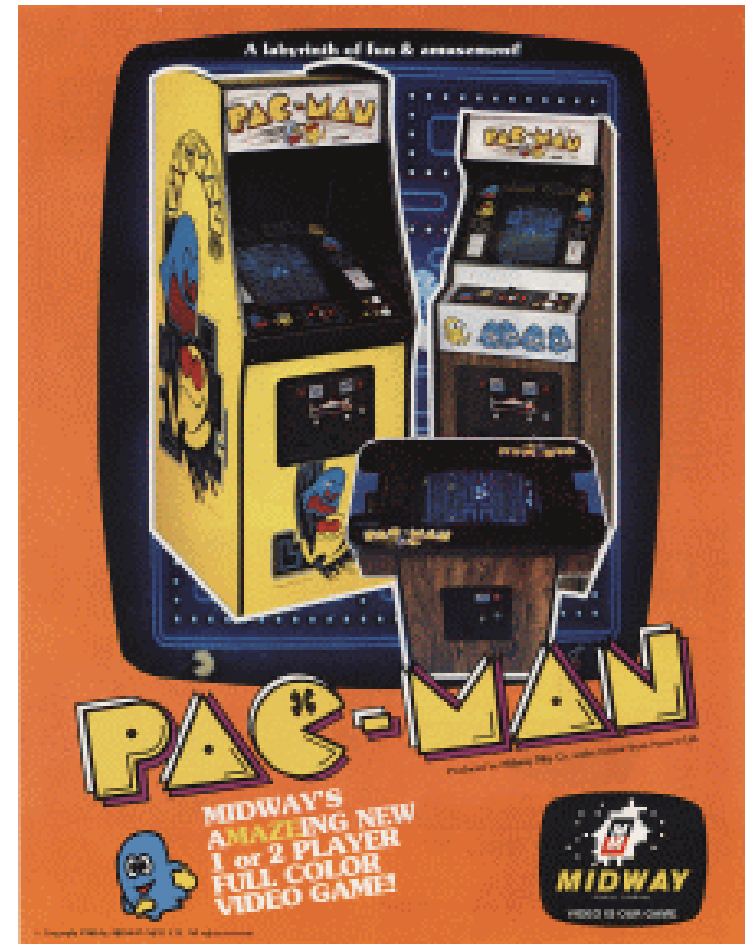
版权过去用于保护计算机程序中具有创造性作者身份的部分

- Copyright used to protect parts of computer program that are **works of creative authorship**, e.g.,
 - source code logic 源代码逻辑
 - Look and feel of whole game (e.g., Pac-Man) 整个游戏的外观和感觉
- Trademark used to protect parts of program that uniquely identify it, and make it distinctive, **for the purposes of commerce** 商标用于保护程序中能独特标识其身份并使其具有区分度的部分，以用于商业目的
 - E.g., BlueNile.com jewelry store sued Ice.com jewelry store for allegedly copying look and feel 例如，BlueNile.com 珠宝店起诉 Ice.com 珠宝店，指控其抄袭外观和感觉
- Notice, both copyright and trademark can be used (in some circumstances) to protect look and feel, e.g., Pac-Man 注意，在某些情况下，版权和商标都可以用于保护外观和感觉，例如《吃豆人》

版权保护软件的“表达”（代码、界面、艺术），商标/Trade Dress 保护软件的“识别性外观”（品牌风格），两者可以共同保护软件的 look & feel。

K.C. Munchkin vs. Pac-Man

- First there was Namco's 1980 arcade game *Pac-man* (see, right)
- Atari exclusively licensed to produce the first play-at-home version of *Pac-Man* (released 1982)
- Philips produced a clone of *Pac-man* (not a direct copy)—*K. C. Munchkin*—released in 1981 (i.e., before Atari's release)
- Atari took Philips to court—won **copyright infringement** on “**look and feel**” of the software



K.C. Munchkin vs. Pac-Man



[https://en.wikipedia.org/wiki/Munchkin_\(video_game\)](https://en.wikipedia.org/wiki/Munchkin_(video_game))

K.C. Munchkin vs. Pac-Man

K.C. Munchkin vs. Pac-Man



K.C. Munchkin!
Magnavox Odyssey²



Pac-Man
Atari 2600

<https://www.youtube.com/watch?v=LXawm3pURjk>

Notes of Software Copyright 软件版权

- Software copyright:
 - Work can be licensed to others 作品可以授权给他人使用
 - Copyright can be sold or assigned to others 版权可以出售或转让给他人
 - Claims for infringement must be proven in court 侵权指控必须在法庭上得到证实
 - Copyright in work done as part of a job belongs to the employer 作为工作的一部分完成的作品版权属于雇主
 - All digital media are likely to be subject to copyright 所有数字媒体都有可能受版权保护：只要是 原创表达（creative expression），就自动受保护
 - Creative commons licenses are like open source licenses for digital media rather than software Creative Commons（CC）是针对数字媒体的“开放许可”，类似软件行业的开源许可。

CC 提供不同层级的授权：

CC 许可证	意义
CC-BY	可以用，但必须署名
CC-BY-SA	必须署名 + 相同方式共享
CC-BY-NC	只能非商业用途
CC0	公共领域，无需署名
CC-BY-ND	可以用但不能改

关键点：

Creative Commons 专注于 媒体内容，而不是软件源代码（软件一般用 MIT/GPL 类许可证）。

软件的专利（争议）

Patents for Software (Critique)

软件和算法专利的使用颇具争议，考虑以下情况：

- Use of patents for software and algorithms controversial, consider:
 - Amazon's *1-click* Patent (novel, but *nonobvious*?) 亚马逊的一键购买专利（新颖，但非显而易见？）
 - Google's *Page Rank* algorithm (novel, but *non-trivial*?) 谷歌的网页排名算法（新颖，但非微不足道？）
- **Criticism 1: Patent Office not competent to decide whether software idea is nonobvious** 批评 1：专利局无能力判定软件创意是否非显而易见
 - This benefits large corporations over small companies / individuals because of prohibitive costs of patent application 这使大公司比小公司/个人更具优势，因为专利申请费用高昂
 - Also, difficult to develop new and innovative software, because hard to know if new software is violating one of thousands of existing patents 此外，开发新的创新软件也变得困难，因为难以知晓新软件是否侵犯了成千上万的现有专利
- **Criticism 2: Using patents to protect an algorithm** 批评 2：用专利保护算法基本数学事实不能申请专利（这会损害公共利益）
 - You cannot patent a *basic mathematical fact* (it would harm public good)
 - Some algorithms can be viewed as basic mathematical facts (the formulas are simple to write down and can be found on Wikipedia), but perhaps they were hard to discover. 某些算法可被视为基本数学事实（公式简单易写且可在维基百科上找到），但或许其发现过程十分艰难。

Notes on Patents for Software 软件专利

- Complex, contentious area 复杂且存在争议的领域
- Legal protection from others exploiting the invention 为防止他人使用发明，提供法律保护
- Use of invention can be licensed to others 发明的使用可以授权给他人
- Typically protected for up to 20 years 通常保护期限可达 20 年
- The essence of the invention is disclosed and published 发明的本质会被披露和公布：专利必须公开其本质内容，公开后所有人都能看到。
- Claims for infringement must be proven in court 对侵权的指控必须在法庭上得到证实
- But there is no agreement 但各方对此并无共识：然而，社会对软件专利是否应该存在，没有共识。



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Summary

Summary

软件到底应该用哪种知识产权保护

- Qu: Which type of intellectual property protection should apply to software?
 - This is a trap! 这是一个陷阱
 - There is no clear or obvious answer 没有清晰明确的答案
 - The experts disagree 专家意见不一
- Possible solution – a new legal framework 新法律框架
 - Difficult to map existing law, designed for tangible goods onto intangible ones
很难将为有形物品设计的现有法律映射到无形物品上。

软件结构复杂，传统 IP（版权、专利、商标）都只能部分保护软件，因此没有明确答案，甚至专家也没有共识，有必要考虑构建新的软件法律框架。

IP Protection	Copyright ©	Patent	Trademark ™
What type of things can be protected?	Creative works (art, music, drama) in a fixed and tangible form	Inventions (processes, machines, etc.) capable of being made or used	Word, phrase, symbol, or item that identifies product, service or corporation
Rights for the creator?	1. reproduce 2. derive new works 3. distribute copies 4. perform publicly	Exclusive rights to make, use, and sell	Exclusive rights to use TM
Rights for the public?	Rights not covered by author rights, plus: 1. Fair use 2. First sale	Cannot “make, use, offer to sell, or sell” invention without patent holder’s permission	Cannot use TM in a way that is likely to cause confusion to customers
How to obtain protection?	Automatic protection, once works appear in a tangible fixed medium. Cost: free	Must apply to Patent Office and meet criteria: • novel & • non-obvious Cost: \$10,000+	Register TM, but must already use mark to represent product or service. Cost: few hundred dollars
How long does protection last?	(Current protection is) lifetime of author + 70 years	Usually 20 years from application	10 years, can be renewed indefinitely

IP 保护类型	版权 Copyright ©	专利 Patent	商标 Trademark ™
可以保护什么？	创意作品（艺术、音乐、戏剧、文学等），必须以固定且可感知的形式存在	发明（流程、机器、制造方法等），必须能够被制造或使用	识别产品、服务或公司的文字、短语、符号、标志、颜色、声音、形状等
创作者拥有的权利？	1. 复制权 (reproduce) 2. 衍生作品权 (改编) 3. 分发复制品 4. 公开表演或展示	独占权：制造、使用、销售发明（或授权他人）	独占使用商标的权利（防止他人使用可能导致混淆的标志）
公众的权利？	除作者权利以外，还包括：1. 合理使用 (Fair Use) 2. 首次销售原则 (First Sale：可转卖作品)	未经许可不得制造、使用、提供销售或销售相关发明	不得以可能造成消费者混淆的方式使用商标
如何获得保护？	自动获得，一旦作品以固定形式出现（写下、录下、保存）。费用：免费	必须向专利局申请并满足条件：• 新颖性 (novel) • 非显而易见性 (non-obvious) 费用：通常 1 万美元以上	必须注册商标，并且在注册前必须已经用于商品或服务。费用：几百美元
保护期限？	作者终身 + 70 年（当前法规）	通常为自申请日起 20 年	每 10 年可续期一次，可无限期续期（必须持续使用）

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专利保护发明功能，申请昂贵，仅 20 年。
商标保护品牌识别，需注册，可无限续期。



Workshop

- Mock IP Court Case.